

Conditioning on the empirical distribution of the data

Another interpretation of conformal prediction is given by treating the dataset as an unordered collection of data points, and then using exchangeability to argue that the test point (X_{n+1}, Y_{n+1}) is equally likely to be any one of these points.

equally likely to be any one of these points. In particular, if each data point $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$ is distinct, and if we observe the unordered set of $n + 1$ values (i.e., we observe the unordered set of data points $\{(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})\}$, but do not know which data value corresponds to which index $i \in [n + 1]$), then the test point (X_{n+1}, Y_{n+1}) is equally likely to be any one of these $n + 1$ values.

To formalize this intuition, given the dataset $\mathcal{D}_{n+1} = ((X_1, Y_1), \dots, (X_{n+1}, Y_{n+1}))$ containing the training data and test point, recalling our notation from Chapter 2 we define its empirical distribution as

$$\hat{P}_{n+1} = \frac{1}{n+1} \sum_{i=1}^{n+1} \delta_{(X_i, Y_i)}.$$

Proof of Theorem 3.2 via the empirical distribution of the data

First, since the score function s is assumed to be symmetric, note that $s(\cdot; \mathcal{D}_{n+1})$ depends on the dataset $\mathcal{D}_{n+1}$ only via the empirical distribution $\hat{P}_{n+1}$. To emphasize this point, we will use the notation $s(\cdot; \hat{P}_{n+1})$ in place of $s(\cdot; \mathcal{D}_{n+1})$ throughout the remainder of this proof. We will also write $Z_i = (X_i, Y_i)$ for the i th data point, and $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$.

Let $h : \mathcal{Z} \rightarrow \mathbb{R}$ be any function. Conditioning on $\hat{P}_{n+1}$, by Proposition 2.2, the test point Z_{n+1} has distribution $\hat{P}_{n+1}$. This implies that, again conditioning on $\hat{P}_{n+1}$, the random variable $h(Z_{n+1})$ has distribution $\frac{1}{n+1} \sum_{i=1}^{n+1} \delta_{h(Z_i)}$. In particular,

$$\mathbb{P} \left(h(Z_{n+1}) \leq \text{Quantile}((h(Z_i))_{i \in [n+1]}; 1 - \alpha) \mid \hat{P}_{n+1} \right) \geq 1 - \alpha,$$

by definition of the quantile. Next, observe that since this calculation is carried out conditionally on $\hat{P}_{n+1}$, the same argument holds for a function h that depends on the empirical distribution $\hat{P}_{n+1}$ as well as on a data point $z \in \mathcal{Z}$ —in particular, we can take $h(z) = s(z; \hat{P}_{n+1})$, to obtain

$$\mathbb{P} \left(s(Z_{n+1}; \hat{P}_{n+1}) \leq \text{Quantile}((s(Z_i; \hat{P}_{n+1}))_{i \in [n+1]}; 1 - \alpha) \mid \hat{P}_{n+1} \right) \geq 1 - \alpha.$$

Since $S_i = s(Z_i; \mathcal{D}_{n+1}) = s(Z_i; \hat{P}_{n+1})$ for each $i \in [n+1]$, by definition, we then have

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \hat{P}_{n+1}) = \mathbb{P}\left(S_{n+1} \leq \text{Quantile}(S_1, \dots, S_{n+1}; 1 - \alpha) \mid \hat{P}_{n+1}\right) \geq 1 - \alpha,$$

where the first step applies (3.5). Marginalizing over $\hat{P}_{n+1}$, we have proved the claim.

Tuning based on a plug-in estimate of the error rate

consider an especially simple and intuitive interpretation of split conformal prediction.

consider prediction sets of the form

$$C_\lambda(X_{n+1}) = \{y : s(X_{n+1}, y) \leq \lambda\},$$

$S \leq \hat{q} = \frac{S_{(k)}}{k}$
 $k = (1-\alpha)(n+1)$

where $\lambda \in \mathbb{R}$ is a parameter controlling the size of the set. Split conformal prediction can be understood as selecting $\lambda = \hat{q} = \text{Quantile}(S_1, \dots, S_n; (1-\alpha)(1+1/n))$. We now explain this choice of λ .

miscoverage ratio

$S_{(k)}$

$$\hat{R}(\lambda) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{Y_i \notin C_\lambda(X_i)\} = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{s(X_i, Y_i) > \lambda\} = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{S_i > \lambda\}$$

If the data points (X_i, Y_i) are i.i.d. draws from some distribution P on $X \times Y$, then $\hat{R}(\lambda)$ is a noisy estimate of the population miscoverage,

$$R(\lambda) = \mathbb{E}_P[\mathbb{1}\{Y \notin C_\lambda(X)\}] = 1 - \mathbb{P}_P(Y \in C_\lambda(X)).$$

$$\begin{aligned} \mathbb{P}_P(Y \in C_\lambda(X)) \\ = 1 - R(\lambda) \geq 1 - \alpha \end{aligned}$$

If we had knowledge of $R(\lambda)$ we would simply set λ to be the smallest value such that $R(\lambda) \leq \alpha$. In words, we take the smallest sets that have the desired coverage level.

we do not know $R(\lambda)$, one reasonable approach is a plug-in method: we could choose $\hat{\lambda}$ to be the smallest value such that $\hat{R}(\lambda) \leq \alpha$.

To correct for this issue, we will now see that we should instead find a value λ such that the coverage is at least $(1-\alpha)(1+1/n)$ on the calibration data. That is, we should choose $\hat{\lambda}$ as

$$\hat{\lambda} = \inf \left\{ \lambda \in \mathbb{R} : \hat{R}(\lambda) \leq \alpha' \right\}, \quad (3.11)$$

where $\alpha' = \alpha - \frac{1-\alpha}{n}$.

$$\hat{R}(\lambda) \leq \alpha' \leq \alpha$$

Proposition 3.10: Split conformal prediction via a plug-in estimate of risk

The split conformal prediction set $\mathcal{C}(X_{n+1})$ defined in (3.8) satisfies

$$\mathcal{C}(X_{n+1}) = \left\{ y \in \mathcal{Y} : s(X_{n+1}, y) \leq \hat{\lambda} \right\},$$

where $\hat{\lambda}$ is defined in (3.11).

Proof of Proposition 3.10

It suffices to prove that $\hat{\lambda}$ is equal to the conformal quantile $\hat{q}$, i.e., the $(1 - \alpha)(1 + 1/n)$ quantile of the calibration scores $s(X_1, Y_1), \dots, s(X_n, Y_n)$. By definition of $\hat{R}$ and of α' , we calculate

$$\begin{aligned} \hat{R}(\lambda) \leq \alpha' &\iff \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{S_i > \lambda\} \leq \alpha - \frac{1 - \alpha}{n} \quad \alpha' \\ &\iff \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{s(X_i, Y_i) \leq \lambda\} \geq 1 - \left(\alpha - \frac{1 - \alpha}{n} \right) = (1 - \alpha)(1 + 1/n). \end{aligned}$$

Plugging this into the definition of $\hat{\lambda}$, we have

$$\hat{\lambda} = \inf \left\{ \lambda \in \mathbb{R} : \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{s(X_i, Y_i) \leq \lambda\} \geq (1 - \alpha)(1 + 1/n) \right\},$$

which is exactly the definition of the conformal quantile $\hat{q}$ for split conformal.

Proof of Theorem 3.2 (split conformal case) via tuning an estimate of error

We will begin by defining an oracle threshold:

$$\lambda^* = \inf \left\{ \lambda \in \mathbb{R} : \frac{1}{n+1} \sum_{i=1}^{n+1} \mathbb{1}\{S_i > \lambda\} \leq \alpha \right\}.$$

The quantity λ^* is an oracle version of $\hat{\lambda}$ that depends on the full dataset $\mathcal{D}_{n+1}$, including both the calibration and test data (while, in contrast, $\hat{\lambda}$ depends only on the calibration data $\mathcal{D}_n$). Since λ^* is a symmetric function of the $n+1$ data points, exchangeability of

the data implies that

$$\mathbb{P}(S_{n+1} > \lambda^*) = \mathbb{P}(S_i > \lambda^*), \quad (3.12)$$

for all $i \in [n+1]$ (recall Lemma 2.3).

Below, we will show that $\lambda^* \leq \hat{\lambda}$ must always hold. This is enough to prove the result, since we then have

$$\begin{aligned} \mathbb{P}(Y_{n+1} \notin \mathcal{C}(X_{n+1})) &= \mathbb{P}(S_{n+1} > \hat{\lambda}) \text{ by Proposition 3.10} \\ &\leq \mathbb{P}(S_{n+1} > \lambda^*) \text{ since } \lambda^* \leq \hat{\lambda} \rightarrow \text{why} \\ &= \frac{1}{n+1} \sum_{i=1}^{n+1} \mathbb{P}(S_i > \lambda^*) \\ &= \mathbb{E} \left[\frac{1}{n+1} \sum_{i=1}^{n+1} \mathbb{1}\{S_i > \lambda^*\} \right] \leq \alpha, \end{aligned}$$

where the third step holds by (3.12), while the last step applies the definition of λ^* .

To conclude, we show that $\lambda^* \leq \hat{\lambda}$. For all $\lambda \in \mathbb{R}$,

$$\begin{aligned} \hat{\lambda} &= \inf \left\{ \lambda : \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{S_i > \lambda\} \leq \alpha' \right\} \Leftrightarrow \frac{1}{n+1} \sum_{i=1}^n \mathbb{1}\{S_i > \lambda\} + \frac{1}{n+1} \leq \alpha \\ \lambda^* &= \inf \left\{ \lambda : \frac{1}{n+1} \sum_{i=1}^{n+1} \mathbb{1}\{S_i > \lambda\} \leq \alpha \right\} \end{aligned}$$

$\alpha - \alpha(\frac{1}{n+1})$ $\lambda^* = \inf \{ \lambda : \frac{1}{n+1} \sum_{i=1}^{n+1} \mathbb{1}\{S_i > \lambda\} \leq \alpha \}$ B

This means the infimum in the definition of λ^* is taken over a larger set than the infimum in the definition of $\hat{\lambda}$, proving the desired claim.

$$\lambda^* \leq \hat{\lambda}$$

$$A \Rightarrow B$$

$$A \leq B$$

$$? \geq \mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha$$

Can conformal prediction be overly conservative?

Theorem 3.11: Bounding the overcoverage of conformal prediction

Under the conditions of Theorem 3.2, we have that

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \leq \frac{\lceil (1 - \alpha)(n + 1) \rceil}{n + 1} + \epsilon_{\text{tie}} \leq 1 - \alpha + \frac{1}{n + 1} + \epsilon_{\text{tie}},$$

where ϵ_{tie} captures the likelihood of the score of the $(n + 1)$ st data point being tied with any other data point,

$$\epsilon_{\text{tie}} = \mathbb{P}(\exists j \in [n], S_{n+1} = S_j).$$

Proof of Theorem 3.11

In our first proof of Theorem 3.2 (given in Section 3.3), we derived an equivalent characterization of the coverage event (see (3.5)):

$$Y_{n+1} \in \mathcal{C}(X_{n+1}) \iff S_{n+1} \leq \text{Quantile}(S_1, \dots, S_{n+1}; 1 - \alpha).$$

Writing $S_{(1)} \leq \dots \leq S_{(n+1)}$ as the order statistics of the scores $S_1, \dots, S_{n+1}$, we can equivalently write

$$Y_{n+1} \in \mathcal{C}(X_{n+1}) \iff S_{n+1} \leq S_{(k)} \text{ where } k = \lceil (1 - \alpha)(n + 1) \rceil, \quad (3.13)$$

by the equivalence between quantiles and order statistics (Fact 2.10). We can assume $k \in [n]$ to avoid the trivial case.

Next, by definition of the order statistics,

$$S_{n+1} \leq S_{(k)} \iff \text{either } S_{n+1} < S_{(k+1)} \text{ or } S_{n+1} = S_{(k)} = S_{(k+1)}.$$

If $S_{n+1} = S_{(k)} = S_{(k+1)}$ then we must have $S_{n+1} = S_j$ for some $j \in [n]$, and therefore,

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) = \mathbb{P}(S_{n+1} < S_{(k+1)}) + \mathbb{P}(S_{n+1} = S_{(k)} = S_{(k+1)}) \leq \mathbb{P}(S_{n+1} < S_{(k+1)}) + \epsilon_{\text{tie}}.$$

Finally, to complete the proof, we have

$$\mathbb{P}(S_{n+1} < S_{(k+1)}) \leq \frac{(k + 1) - 1}{n + 1} = \frac{\lceil (1 - \alpha)(n + 1) \rceil}{n + 1}$$

by Fact 2.15(i)

$$\mathbb{P}(S_i < S_{(k)}) \leq \frac{k-1}{n}$$

$$\sum_{i \in I} = 0$$

Proposition 3.12

Under the conditions of Theorem 3.2, if the scores $S_1, \dots, S_{n+1}$ are distinct almost surely, then

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) = \frac{\lceil (1-\alpha)(n+1) \rceil}{n+1} \leq 1 - \alpha + \frac{1}{n+1}.$$

$$n \rightarrow \infty$$

Proof of Proposition 3.12

We return to the calculation (3.13). When the scores are distinct with probability 1, exchangeability implies that $P(S_{n+1} \leq S_{(k)}) = \frac{k}{n+1} = \frac{\lceil (1-\alpha)(n+1) \rceil}{n+1}$ (by Fact 2.15(iv)), as desired.

$$\{y: S_1 \leq S_{(k)}\}$$

$$\uparrow \quad \downarrow$$

$$|y_i - f(x_i)|$$

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid X_{n+1} = x) \geq 1 - \alpha$$

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha$$

$$\uparrow \quad \uparrow \quad \uparrow$$

$$RV \quad \underline{D_{cal}}, \underline{D_{pre}} \quad RV \quad \underline{D_{test}}$$

Conditional Coverage

We first examine the coverage of split conformal prediction conditional on the training set:

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n).$$

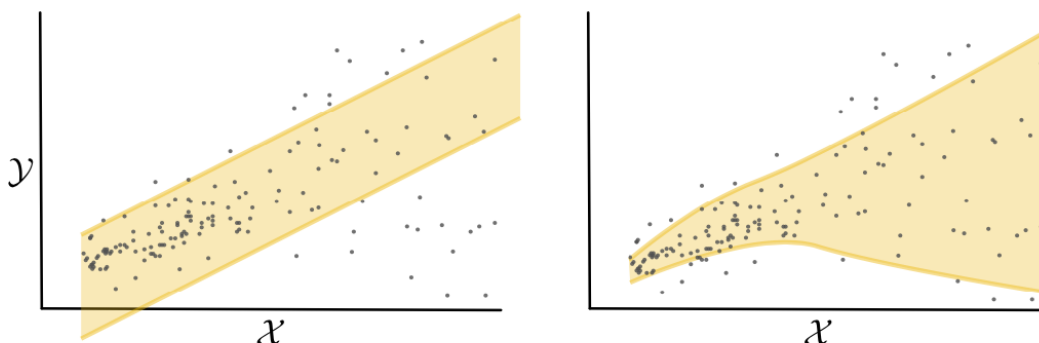
This is the empirical coverage that we would obtain if we used the points $\mathcal{D}_n = ((X_1, Y_1), \dots, (X_n, Y_n))$ for calibration, and then evaluated our coverage over an infinite test set of i.i.d. data. This quantity is a random variable, since it depends on the calibration set $\mathcal{D}_n$.

If the conformal prediction set satisfies marginal coverage at level $1 - \alpha$, then the training-conditional coverage must be $\geq 1 - \alpha$ in expectation:

$$\mathbb{E}[\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n)] = \mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha.$$

However, a marginal coverage guarantee does not ensure that training-conditional coverage will be close to $1 - \alpha$ with high probability; for this, we need a stronger concentration-type statement.

46



In fact, there is a closed-form expression that characterizes the training-conditional coverage in this setting. For intuition, let us first consider a simplified setting where the score distribution is continuous (so that there are no ties, almost surely), and where $(1 - \alpha)(n + 1)$ is an integer. In that case, training-conditional coverage is exactly distributed as a Beta distribution:

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n) \sim \text{Beta}((1 - \alpha)(n + 1), \alpha(n + 1)).$$

The Beta distribution arises from the order statistics of the uniform distribution—conformal prediction relies only on the ranks of the scores, so if there are no ties one can without loss of generality analyze conformal scores that are uniformly distributed.

$$S_1, \dots, S_n \mid \mathcal{D}_n \sim \hat{P}_n \quad S_{(1)}, \dots, S_{(n)}$$

Theorem 4.1: Training-conditional coverage for split conformal prediction

Suppose the data points (X_i, Y_i) are i.i.d., and let $\mathcal{C}(X_{n+1})$ be constructed via split conformal prediction (Algorithm 3.6) using any pretrained score function s . Then the training-conditional coverage $\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n)$ stochastically dominates the $\text{Beta}((1 - \alpha)(n + 1), \alpha(n + 1))$ distribution, and in particular, for any $\Delta \geq 0$,

$$\mathbb{P}\left(\underbrace{\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n)}_{\geq 1 - \alpha} \leq 1 - \alpha - \Delta\right) \leq F_{\text{Beta}((1-\alpha)(n+1), \alpha(n+1))}(1 - \alpha - \Delta) \leq e^{-2n\Delta^2},$$

where $F_{\text{Beta}(a,b)}$ denotes the CDF of the $\text{Beta}(a, b)$ distribution.

$$\geq 1 - \alpha$$

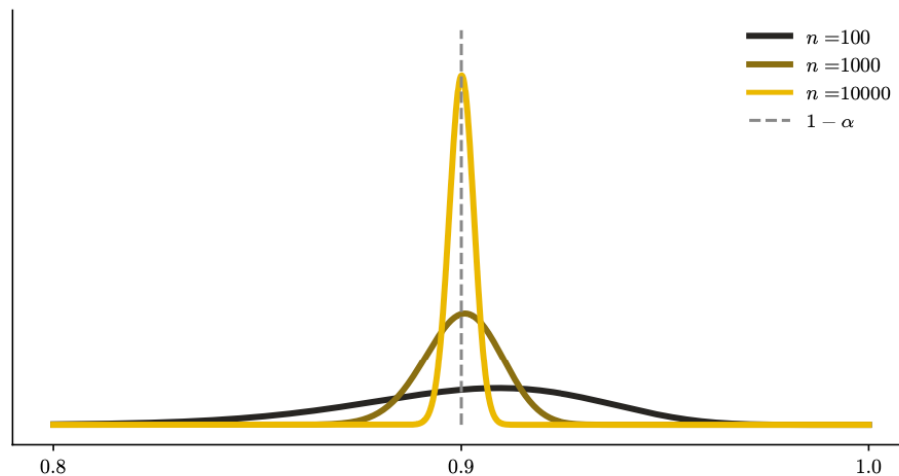


Figure 4.2: The distribution of coverage of split conformal prediction conditional on the training data, when the scores have no ties. The distribution concentrates around $1 - \alpha$ with rate $n^{-1/2}$.

4.1.1 High-probability coverage

As described above, the marginal coverage property is equivalent to a bound on the *expected* training-conditional miscoverage,

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha \iff \mathbb{E}[\mathbb{P}(Y_{n+1} \notin \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n)] \leq \alpha,$$

where the expectation is over the calibration data $\mathcal{D}_n$. We can also consider a different approach: instead of bounding the expected error, we might wish to bound the error with some desired probability—that is, in this setting, we enforce a bound on the chance that we end up with poor training-conditional coverage. Concretely, for some $\delta \in (0, 1)$, we could ask for a prediction set $\mathcal{C}(X_{n+1})$ that satisfies

$$\mathbb{P}(\mathbb{P}(Y_{n+1} \notin \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n) \leq \alpha) \geq 1 - \delta. \quad (4.1)$$

In other words, there is only a δ chance that we draw a calibration set $\mathcal{D}_n$ leading to a miscoverage rate on future data that is higher than α .

In the special case of i.i.d. data, we can produce sets $\mathcal{C}(X_{n+1})$ that satisfy the error-control property in (4.1), with a slight modification of split conformal prediction. Taking as a starting point the distribution of training-conditional coverage given in Theorem 4.1, we see that by defining

$$\mathcal{C}(X_{n+1}) = \{y \in \mathcal{Y} : s(X_{n+1}, y) \leq \text{Quantile}(S_1, \dots, S_n; (1 - \alpha')(1 + 1/n))\}$$

(i.e., split conformal run at a nominal level $1 - \alpha'$), then we will achieve the desired high probability bound (4.1) if we choose the appropriate nominal level $1 - \alpha'$. In particular, given a fixed $\delta > 0$, we can take α' to be the unique value that satisfies

$$F_{\text{Beta}((1-\alpha')(n+1), \alpha'(n+1))}(1 - \alpha) = \delta.$$

By our calculations on the mean and variance of the Beta distribution, above, we can see that $\alpha' = \alpha - \mathcal{O}(n^{-1/2})$ —that is, this is simply a slightly more conservative version of split conformal prediction.

$F(\cdot)$ is cdf of S_{n+1} or S

Proof of Theorem 4.1

Throughout this proof, we will treat the pretrained score function as fixed—formally, recalling Algorithm 3.6, we are conditioning on the pretraining set $\mathcal{D}_{\text{pre}}$.

Let F be the CDF of the distribution of $s(X, Y)$, when the data point is sampled as $(X, Y) \sim P$. Define $S_i = s(X_i, Y_i)$ for $i \in [n+1]$, which are i.i.d. draws from the distribution with CDF F . Next, let $S_{(1)} \leq \dots \leq S_{(n)}$ be the order statistics of $S_1, \dots, S_n$. By definition of split conformal, we have $Y_{n+1} \in \mathcal{C}(X_{n+1})$ if and only if $S_{n+1} \leq S_{(k)}$ for $k = \lceil (1 - \alpha)(n+1) \rceil$, and so we have

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n) = \mathbb{P}(S_{n+1} \leq S_{(k)} \mid \mathcal{D}_n) = F(S_{(k)}),$$

where the last step holds since S_{n+1} is independent of $\mathcal{D}_n$ and has CDF F , while $S_{(k)}$ is a function of $\mathcal{D}_n$. Therefore,

$$\mathbb{P}(\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n) \leq 1 - \alpha - \Delta) = \mathbb{P}(F(S_{(k)}) \leq 1 - \alpha - \Delta).$$

Next let $U_i = F(S_i)$ for all $i \in [n]$. Then the U_i 's are i.i.d., and by the basic property of CDFs (2.7), their shared distribution is superuniform—that is, $\mathbb{P}(U_i \leq \tau) \leq \tau$ for all $\tau \in [0, 1]$. Note also that, by monotonicity of F , the order statistics $U_{(1)} \leq \dots \leq U_{(n)}$ of $U_1, \dots, U_n$ satisfy $U_{(k)} = F(S_{(k)})$ (see Fact 2.13(i)). Then

$$\mathbb{P}(F(S_{(k)}) \leq 1 - \alpha - \Delta) = \mathbb{P}(U_{(k)} \leq 1 - \alpha - \Delta) \leq \mathbb{P}(U_{(k)}^* \leq 1 - \alpha - \Delta),$$

where $U_1^*, \dots, U_n^*$ are i.i.d. $\text{Unif}[0, 1]$ random variables, with order statistics $U_{(1)}^* \leq \dots \leq U_{(n)}^*$. Finally, by definition of the Beta distribution, the k th order statistic from n i.i.d. uniform random variables has distribution

$$U_{(k)}^* \sim \text{Beta}(k, n+1-k).$$

This distribution stochastically dominates the $\text{Beta}((1 - \alpha)(n+1), \alpha(n+1))$ distribution, since $k \geq (1 - \alpha)(n+1)$. This proves the first inequality claimed in the theorem.

The second inequality (i.e., the bound on the CDF of the Beta distribution) is due to the fact that the $\text{Beta}((1 - \alpha)(n+1), \alpha(n+1))$ distribution has mean $1 - \alpha$, and is $\frac{1}{4n}$ -subgaussian.